

BERT: Encoder-Only Transformer

Deep Learning

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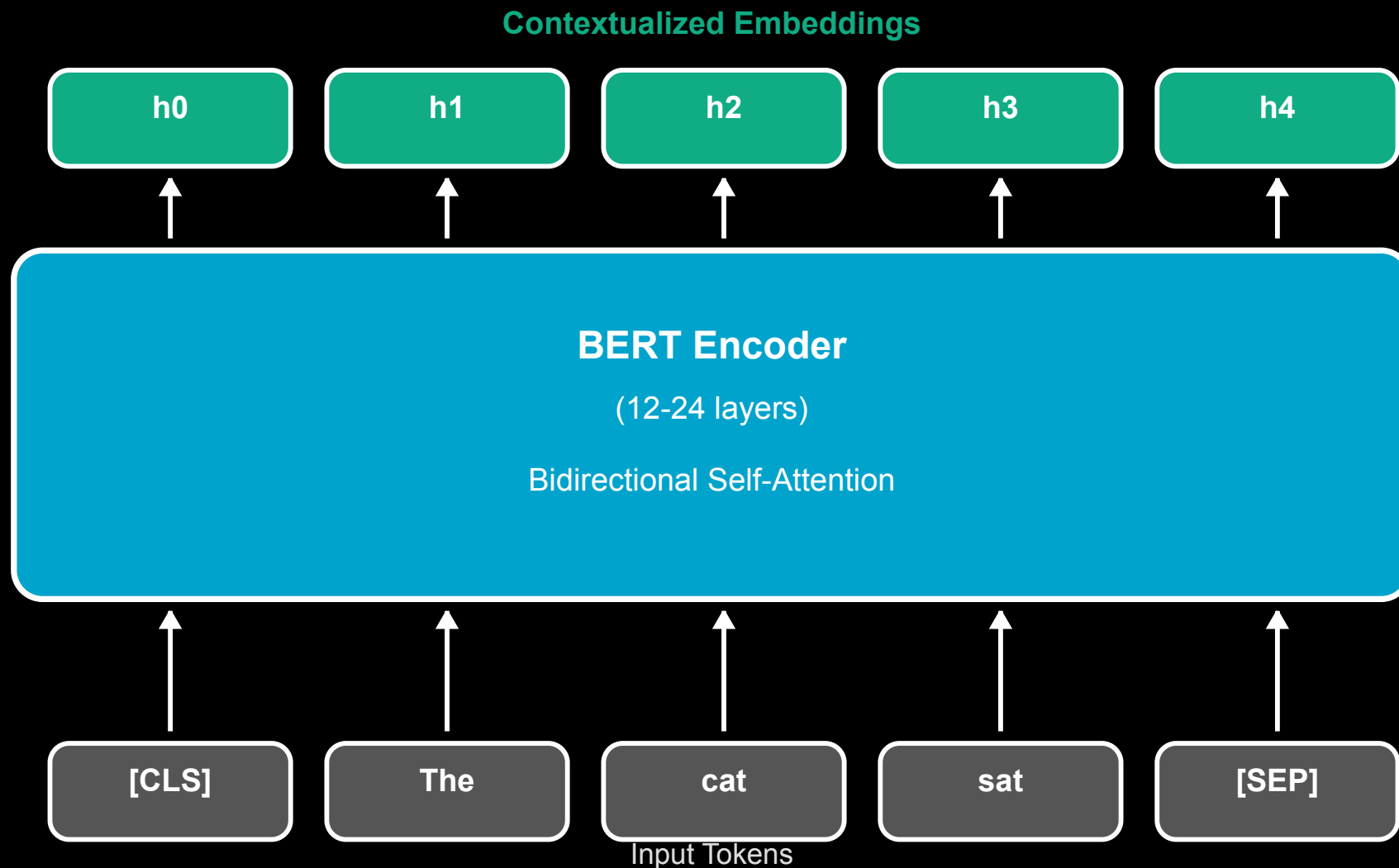
Contents of This Video

In this video, we will cover:

- BERT: encoder-only architecture
- Bidirectional understanding
- Masked language modeling (MLM)
- The [CLS] token for classification
- Fine-tuning BERT for downstream tasks

BERT: Encoder-Only

BERT: Encoder-Only Architecture



Bidirectional Context

BERT sees both directions:

“The cat sat on the [MASK]”

- [MASK] can see “The cat sat on” (left context)
- [MASK] can see nothing to the right
- But “sat” can see BOTH “cat” and “[MASK]”

Contrast with GPT (left-to-right only):

- Each position only sees previous tokens
- Can’t use future context for understanding

Training: Masked Language Modeling

Randomly mask 15% of tokens, predict them:

Input: “The [MASK] sat on the mat”

Output: Predict “cat” at position 2

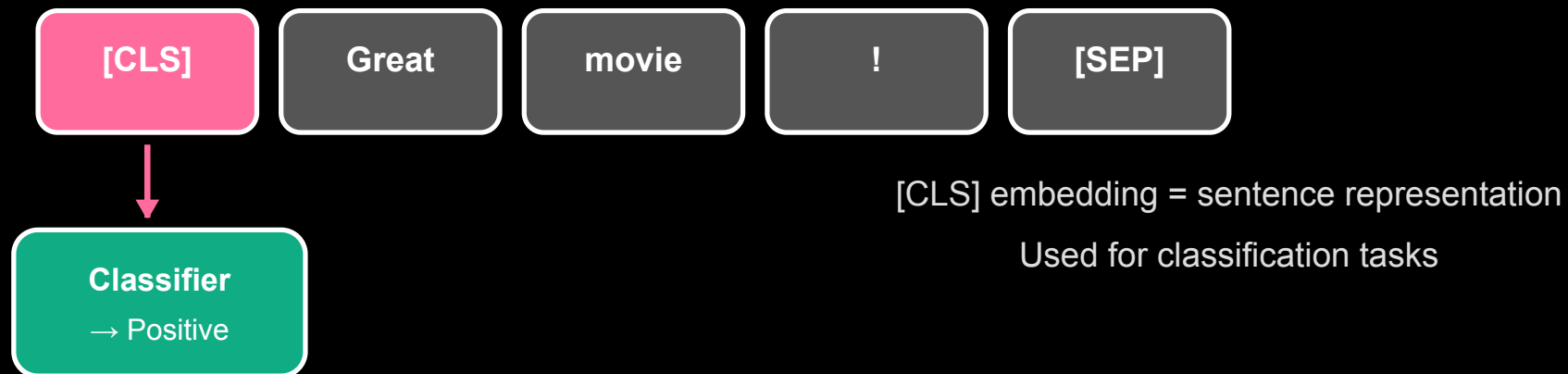
Why masking?

- Can’t use standard language modeling (bidirectional!)
- Masking creates a prediction task
- Forces model to learn deep contextual representations

The [CLS] Token

Special token at position 0:

[CLS] Token: Sentence-Level Representation



Fine-Tuning BERT

Pre-train once, fine-tune for many tasks:

Task	Fine-tuning approach
Sentiment	Classifier on [CLS]
NER	Classifier on each token
Q&A	Start/end token prediction
Similarity	Compare [CLS] embeddings

Transfer learning for NLP!

What We've Covered

In this video, we've learned:

- BERT: encoder-only Transformer (no generation)
- Bidirectional attention: sees all context
- Masked Language Modeling: predict masked tokens
- [CLS] token: sentence-level representation
- Fine-tuning: adapt pre-trained BERT to specific tasks
- BERT pioneered pre-train/fine-tune paradigm in NLP